OPERATIONAL PROCESS MAP & GOVERNANCE DOSSIER

Warehouse Cold Chain & Inventory Management

Operational Process Map

Scope: Governs generic warehouse cold-chain maintenance, thermal envelope integrity and FEFO inventory rotation across all stored seafood commodities.

Warehouse Dynamic Governance

Process Name	Warehouse Cold Storage & Inventory Management
Master Value Chain Stage	Process 3: Warehouse Cold Storage & Processing
Legacy SOP Replacement	Replaces SOP: Cold Chain & Inventory Management
Integrated Management System	ISO 9001:2015 | ISO 14001:2015 | ISO 45001:2018

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Preparation, Review & Approval

Responsibility	Name / Designation	Role
Prepared By	C. T. Wan	Warehouse Cold Storage Supervisor
Reviewed By	K. W. Chan	Logistics & Warehouse Operations Lead
Approved By	Quality Control Manager (Horizontal RCA Team Lead) & VP of Operations	

Document Structure

Part	Section	Purpose
Part 1	Operational Process Map (OPM)	Process boundaries, workflow, QHSE risks, operational controls, KPIs and improvement triggers.
Part 2	Process Card Register (PCR)	Order-specific risks, controls, execution records and release requirements.
Part 3	Static Evidence	Proof of effective end-to-end controls over the risk based operational steps and stated process goals.

Stage 1 — Scoping Boundaries & Horizontal Auditing Interfaces

INBOUND INTERFACE Entry Gate	PROCESS EXECUTION CORE	OUTBOUND INTERFACE Exit Gate
Upstream source Process 2 — Inbound Receiving & Inspection / Facility Put-Away. **Entry requirements** • Approved Material Receiving Inspection Report (MRIR). • Verified receiving core temperature. • Intact vacuum/brine packaging. • WMS barcode and FEFO label mapped.	**Activities covered** • Cargo ingestion. • Chilled and deep-freeze storage. • Thermal monitoring. • FEFO auditing. • Cycle counting. • Pick-release. **Facilities covered** Tsing Yi Logistics Hub — Cold Storage Zone B and Zone F.	**Downstream receiver** Process 4 — Order Management & Fulfillment / Order Processing & Picking. **Clean handshake** • Validated FEFO pick-list. • Verified product temperature ≤ +4.0°C. • Packaging and saturation media intact. • WMS barcoded electronic transfer with zero discrepancies.

Interface Rule	Detailed Requirement
Inbound Prerequisite	Cargo must have a validated MRIR, verified receiving core temperature of **0°C to +4.0°C for chilled items** or **≤ −18.0°C for frozen items**, zero signs of vacuum/packaging puncture, and active FEFO batch barcodes generated by the ERP/WMS.
Outbound Clean Handshake	Product must be picked strictly according to the FEFO priority queue. Verify the stated core/surface temperature requirement of **0°C to +4.0°C**, intact packaging and saturation media, and confirmed digital WMS inventory transfer without physical variance, thermal damage or stock confusion.

Stage 2 — Activity Mapping & Operational Sequence

The workflow uses three functional swimlanes and six sequential macro-steps.

Responsible Swimlane	Process Flow
Warehouse Operator	**3.1 Ingest Cargo & Scan WMS Barcode** ↓ **3.2 Transfer to Dedicated Thermal Zone** ↓ **3.3 Execute FEFO Slotting & Rack Placement**
Automated IoT & WMS Controller	**3.4 Continuous Thermal Envelope Surveillance** **Decision: Temperature ≤ +4.0°C and saline intact?** **YES → Maintain active holding cycle** **NO → Trigger HOLD / QUARANTINE sub-routine** **Route to isolated chilled hold room**
Warehouse QA Officer / Supervisor	**3.5 Daily FEFO Audit & Product Integrity Cycle Check** ↓ **3.6 Authorize Pick-Release** ↓ **Clean Outbound Handshake to Process 4**

Detailed Work Instructions

Step	Activity	Execution Requirements
3.1	**Ingest Cargo & Scan WMS Barcode**	Verify physical pallet barcodes against the incoming Inbound Receiving Record. Confirm batch/lot, harvest/processing date and system-calculated FEFO expiration date before opening storage access gates.
3.2	**Transfer to Dedicated Thermal Zone**	Transport palletized cargo using certified electric pallet trucks into: • **Zone B — Chilled:** 0°C to +4.0°C. • **Zone F — Frozen:** ≤ −18°C. Complete physical staging transition within **≤ 10 minutes** to prevent surface condensation.
3.3	**Execute FEFO Slotting & Rack Placement**	Slot cargo into assigned gravity-fed or multi-tier rack bays. Physically align earlier-expiring stock at the front pick-face and newer inbound stock behind, following First-Expired, First-Out.
3.4	**Continuous Thermal Envelope Surveillance**	Maintain uninterrupted automated refrigeration control using wireless IoT sensors logging every **5 minutes**. **Closed-loop decision gate:** If storage temperature is **> +4.0°C for > 15 minutes** or **> +6.0°C instantaneously**: • Trigger acoustic/visual Andon alarm. • Automatically engage the secondary cooling loop. • Route the batch to physical quarantine.
3.5	**Daily FEFO Audit & Product Integrity Cycle Check**	Inspect stock daily for expiration dates, pallet integrity, carton crushing, vacuum loss and brine seepage. Calibrate IoT nodes against the NIST master thermometer.
3.6	**Authorize Pick-Release to Order Fulfillment**	System-validate and sign off WMS electronic pick waves. Ensure the picking queue releases the earliest-expiring batch. No manual override is permitted unless authorized by QA.

Stage 3 — Integrated Risk Assessment: QHSE

Evaluation scale: Consequence Severity (S: 1–5) × Likelihood (L: 1–5) = Risk Priority Score (RPS: 1–25).
Mandatory engineering-control rule: Any RPS ≥ 15 or severity S = 5 requires an engineered in-process control.

Step	Activity / Lens	Hazard / Failure Mode	S	L	RPS	Priority
3.1	**Ingest Cargo & Scan Barcode** Quality	Inaccurate expiration date entered into WMS, causing FEFO bypass and delivery of expired seafood to clients.	4	2	8	Medium
3.2	**Transfer to Thermal Zone** Environment	Excessive loading-bay door-open duration, causing thermal air leakage and high chiller power draw.	2	4	8	Medium
3.2	**Transfer to Thermal Zone** Health & Safety	Slip-and-fall hazard from condensed floor moisture or leaked brine at doorway transition zones.	3	3	9	Medium
3.3	**Execute FEFO Slotting** Quality	Physical stock misplacement, with new stock in front of old stock, causing batch obsolescence and bacterial spoilage.	4	3	10	Medium
3.3	**Execute FEFO Slotting** Health & Safety	Musculoskeletal injury from manually lifting heavy master cartons (> 18 kg), or injury from falling pallets from rack bays.	3	3	9	Medium
3.4	**Thermal Surveillance** Quality	Chiller failure or setpoint drift (> +4.0°C; critical limit > +6.0°C), driving rapid Vibrio parahaemolyticus / histamine proliferation.	5	3	**15**	**HIGH**

Step	Activity / Lens	Hazard / Failure Mode	S	L	RPS	Priority
3.4	**Thermal Surveillance** Environment	Catastrophic fluorocarbon (HFC/R404A) refrigerant leak exceeding statutory limits into the atmosphere.	4	2	8	Medium
3.5	**Daily Stock Integrity Audit** Quality	Undetected packaging vacuum loss or saline desaturation, causing flesh mushiness, aerobic bacterial decay and food poisoning.	5	3	**15**	**HIGH**
3.6	**Pick-Release Authorization** Quality	Operator manually overrides the picking queue, releasing a non-conforming, warm or expired lot to supermarket clients.	5	3	**15**	**HIGH**

Risk Control Dimension	Profile A: Standard Chilled / Frozen Whole Fish
Example Product	Chilled or frozen whole fish
Storage Temperature Band	**Chilled:** 0.0°C to +4.0°C **Frozen:** ≤ −18.0°C
Thermal Action Limit	Temperature > +5.0°C for > 30 minutes triggers review.
Slotting & Access Rule	Standard static racking; monthly cycle count.
Media / Brine Monitoring	Visual ice-pack check.
Release & Authorization Interlock	Hard digital interlock: blocks manual overrides and requires barcode scanning to verify products before release,

Stage 4 — Operational In-Process Controls & Dynamic Profiles

4.1 — 6M Input Controls: Pre-Execution Gate

6M Dimension	Non-Negotiable Operational Requirement
1. Man	Certified cold-chain warehouse specialist; cold-environment PPE compliance (ISO 45001).
2. Machine	Serviced electric reach truck / pallet stacker; fast-acting high-speed roll-up doors.
3. Material	Inbound seafood in intact vacuum packaging and sealed master cartons with barcode tags.
4. Method	WMS-directed slotting rules enforcing strict FEFO logic; zero manual location bypass.
5. Measurement	Calibrated dual-probe wireless IoT sensors, ±0.1°C accuracy, with battery health > 80%.
6. Milieu	Zone B chilled warehouse envelope: 0.0°C to +4.0°C; relative humidity 85%–90%; clean and dry floor.

4.2 — Mandatory In-Process Controls

Category	Control	Requirement / Response
A — Process Criteria **Execution Guardrails**	**Refrigeration Interlock**	Fast-acting automatic air curtains interlocked with dock doors. Maximum open-cycle duration ≤ 45 seconds to eliminate thermal plume loss.
	Poka-Yoke Racking	Gravity-fed roller racks for Profile B items. Pallets load from the rear and roll forward to the pick-face, mechanically preventing newer stock from being picked ahead of older stock.
B — Monitoring & Stop Authority	**Real-Time IoT Telemetry**	Multi-sensor ambient mapping at upper, middle and floor tiers, with readings logged every 5 minutes.

Category	Control	Requirement / Response
Surveillance	**Mandatory Deviation Stop Rule**	If Zone B temperature exceeds **+4.0°C for ≥ 15 minutes**, or any sensor reads **> +6.0°C**: • Trigger Andon visual alarm. • Send SMS alert to the Warehouse Supervisor and QA Manager. • Automatically freeze the affected WMS storage bin to block outbound picking.
C — Change & Abnormality Management **Operational Resilience**	**Refrigeration Redundancy Switch**	Dual independent condensing compressors with automatic failover in the event of primary motor breakdown.
	Refrigerant Leak Containment	Fixed infrared halogen leak detectors set at 50 ppm trigger emergency exhaust ventilation and shut off liquid-line solenoid valves to prevent atmospheric loss.

Stage 5 — Outcome Controls & Process Effectiveness

Lagging performance indicators

Dimension	Evaluation / KPI	Calculation / Verification Method	Target
Quality	Thermal envelope stability and zero spoilage **Cold Chain Storage Yield**	(Total storage hours within 0°C–4°C ÷ Total recorded hours) × 100	**≥ 99.9%** compliant
Quality	Strict FEFO compliance **FEFO Execution Rate**	(Picks executed in correct FEFO sequence ÷ Total outbound picks) × 100	**100.0%** Zero bypass
Safety	Cold-storage personnel safety **Recordable Incident Rate (RIR)**	Σ (Lost-time slip/trip incidents + Cold-stress events)	**0 incidents**
Environment	Energy and refrigerant stewardship **Refrigerant Loss / Energy Index**	Σ F-gas leaks (kg) + (kWh consumed ÷ Tonnes of cargo stored)	**Zero loss** **≤ 45 kWh/t**
Horizontal Handoff	Quality of inventory released downstream **Clean Handshake Acceptance Rate**	(Batches accepted by Picking Team without deviation ÷ Total batches) × 100	**100.0%** clean handoff

Stage 6 — Opportunity Triggers & Continual Improvement Escalation

Trigger / Review	Condition	Required Escalation / Action
Efficiency Trigger	Manual daily physical cycle counting requires > 2.5 operator hours/day.	Escalate to the IMS Committee for capital investment in automated overhead RFID/BLE shelf-scanning transponders.
Environmental / Energy Trigger	Condenser waste heat is currently vented to the atmosphere.	Evaluate a plate heat exchanger to preheat sanitation wash-water as an ISO 14001 carbon-reduction initiative.
Governance Review	Quarterly consolidated opportunity and temperature-variance review.	Review under ISO 9001 / ISO 14001 / ISO 45001 Clause 9.3 Management Review.